

```
#include <algorithm>

int main() {
    std::vector<int> numbers = {1, 2, 3, 4, 5};

    // Use a lambda expression to print each number
    std::for_each(numbers.begin(), numbers.end(), [](int n) {
        std::cout << n << " ";
    });
    std::cout << std::endl;

    return 0;
}
```

Standard Template Library (STL):

The STL is a powerful library that provides a set of generic classes and functions for data structures and algorithms.

```
#include <iostream>
#include <vector>
#include <algorithm>

int main() {
    std::vector<int> numbers = {5, 3, 1, 4, 2};

    // Sort the numbers
    std::sort(numbers.begin(), numbers.end());

    // Print the sorted numbers
    for (int n : numbers) {
        std::cout << n << " ";
    }
    std::cout << std::endl;

    return 0;
}
```

Industry Trends and Applications

C++ continues to be a dominant language in various industries due to its performance and flexibility. Here are some current trends and applications of C++:

Game Development:

C++ is the language of choice for game development due to its performance and control over hardware. Major game engines like Unreal Engine and Unity use C++ for their core systems.

System Programming:

C++ is extensively used for system programming, including operating systems, device drivers, and embedded systems. Its ability to manipulate hardware directly makes it ideal for these low-level applications.

Finance:

In the finance industry, C++ is used for high-frequency trading and financial modeling applications. Its performance and efficiency are crucial for handling large volumes of data and executing complex algorithms quickly.

Real-Time Systems:

C++ is well-suited for real-time systems where performance and deterministic behavior are critical. It is used in applications such as avionics, telecommunications, and robotics.

Scientific Computing:

C++ is used in scientific computing for simulations, modeling, and computational research. Libraries like Boost and Eigen provide powerful tools for numerical computations and data analysis.

Cloud Computing:

With the rise of cloud computing, C++ is being used to develop high-performance backend services and frameworks that power cloud infrastructures. Its efficiency helps in optimizing resource usage and improving scalability.

Best Practices for C++ Development



To ensure the quality and maintainability of your C++ code, it is important to follow best practices:

1. Use Modern C++:

- Take advantage of modern C++ features introduced in C++11, C++14, C++17, and C++20. These features improve code readability, safety, and performance.